

Quick Check

Lesson 6: Solving Right-Angled Triangles Using Trigonometric Ratios

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 6

A ramp at Kisumu County Referral Hospital makes a 12° angle with the ground. The horizontal distance along the ground is 6 m. You want the length of the ramp surface. Which ratio should you use?

- A Cosine, because the known side is adjacent to 12° and the ramp is the hypotenuse
- B Sine, because the ramp is the side opposite the 12° angle
- C Tangent, because the ground distance is the side opposite the 12° angle
- D Tangent, because a ramp is a slope

A telecom mast in Garissa is 45 m tall and stands at a right angle to the ground. A support wire is anchored on the ground 45 m from the base of the mast. What angle does the wire make with the ground?

A 45°

B 30°

C 60°

D 90°

A student stands 40 m from the base of the Nakuru transmission mast. The angle of elevation to the top is 50° . Using $\tan 50^\circ \approx 1.1918$, what is the height of the mast?

A 47.7 m

B 33.6 m

C 30.6 m

D 25.7 m

A ramp makes a 12° angle with the ground. The horizontal distance along the ground is 6 m. Using $\cos 12^\circ \approx 0.9781$, what is the length of the ramp surface, to 2 decimal places?

A 1.28 m

B 6.13 m

C 28.88 m

D 5.87 m

A ramp at Mombasa port makes a 15° angle with the horizontal. The ramp is 8 m long. What is the vertical height the ramp rises, to 2 decimal places?

A 2.07 m

B 2.14 m

C 30.90 m

D 7.73 m

A surveyor in Garissa measures that a telecom tower casts a shadow 32 m long on flat ground when the sun's angle of elevation is 62° . What is the height of the tower, to 1 decimal place?

A 17.0 m

B 28.3 m

C 15.0 m

D 60.2 m

After solving a right-angled triangle, a learner writes: 'I know one acute angle and one side. Now I can find the other two sides and the other angle.' What does 'solving a right-angled triangle completely' mean?

- A** Finding every unknown side and angle from one known side and one acute angle
- B** Finding the height of the triangle only
- C** Finding only the hypotenuse
- D** Finding the three ratios sin, cos and tan for the 90° angle

For a named acute angle in a right-angled triangle, you know the length of the side opposite it and you want the length of the hypotenuse. Which rule from SOH-CAH-TOA should you use?

- A** SOH, because it uses opposite and hypotenuse
- B** CAH, because it uses the adjacent side
- C** TOA, because it uses the opposite side
- D** TOA, because tangent is the ratio to use whenever an angle is given

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).